

ISSUE 128 · MACHINE-CERTIFIED

4.1357418449×
resource reduction

393 → 95 steps

11,791 → 2,851 merged groups

Same problem. Same circuit. Tighter proof.

Pinned comparison

Every physical and compilation choice is held fixed; only the proof is tightened.

Constraint	Published	Certified	Same?	Evidence
Model	Heisenberg	Heisenberg	YES	$(XX+YY+ZZ)/4$
Lattice	12×12	12×12	YES	periodic, $N=144$
Time	$T=1$	$T=1$	YES	exact rational
Tolerance	$1e-6$	$1e-6$	YES	operator norm
Formula	Suzuki-4	Suzuki-4	YES	five-copy
Per step	31 stages	31 stages	YES	four matchings
Merge rule	$30r+1$	$30r+1$	YES	same compiler
Steps	393	95	—	proof tightened

The gain comes from delaying norm inequalities

OLD PROOF

Expand early
Take absolute values
term by term;
cancellations
disappear.

LOCAL LOG

Merge first
Keep exact algebra,
canonicalize
translations, combine
equal Paulis.

CERTIFY

Close rigorously
Anticommuting groups
plus an outward
rational tail.

Two Pauli sidecars carry the tightening

Proof-carrying objects

Each payload is SHA-256-bound; coverage, support and pairwise anticommutation are independently checked.

75,324

D4 terms
7,576 groups

605,832

D5 terms
123,106 groups

≤ 10

maximum
group size

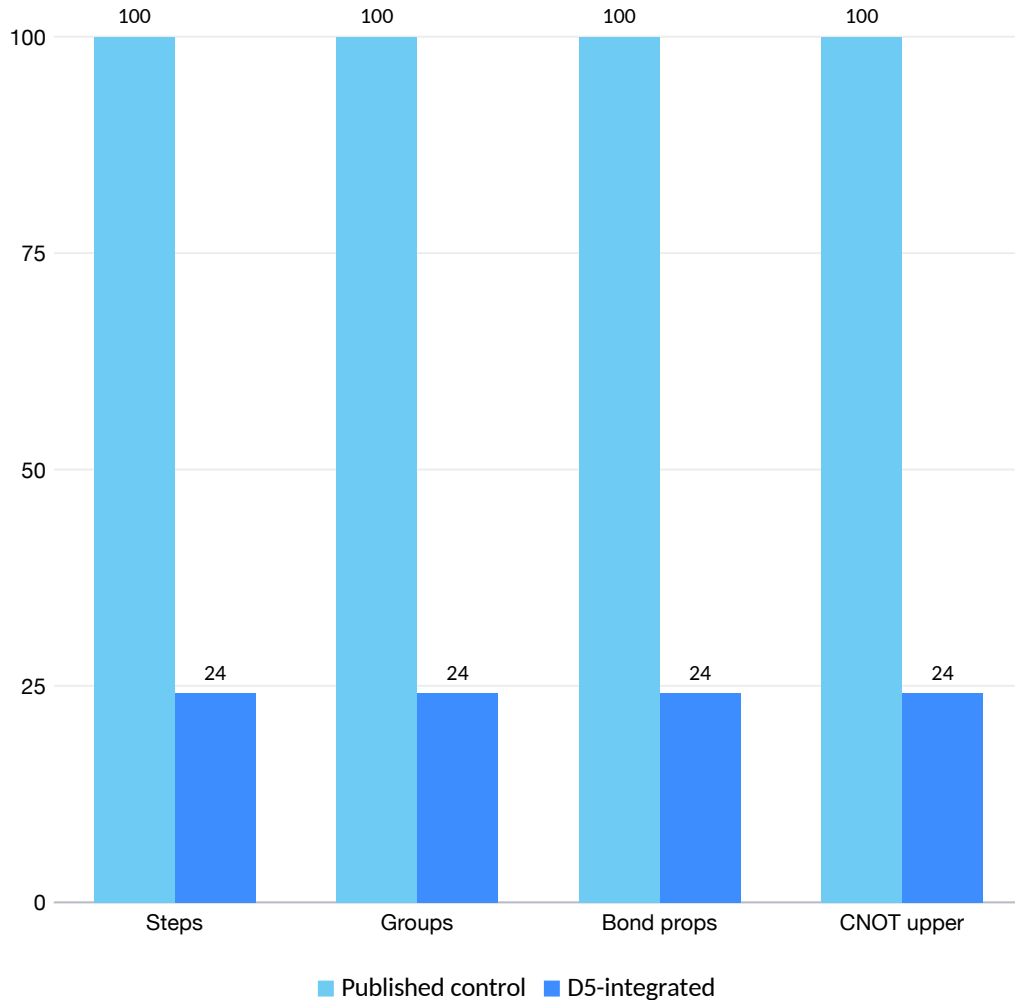
95 passes; 94 fails under the same exact ledger

Exact-rational decision

Displayed decimals are outward-readable projections; acceptance uses integer cross-multiplication.

Ledger item	r=95 upper	Scale	Trust object	Status
D4	5.7218801e-7	r^{-4}	grouped sidecar	checked
D5	3.4853485e-8	r^{-5}	grouped sidecar	checked
D6	1.7893141e-7	r^{-6}	majorant	checked
D7	2.1921361e-8	r^{-7}	majorant	checked
D8+ tail	1.8255488e-7	finite tail	rational	checked
TOTAL @95	9.9044914e-7	$\leq 1e-6$	exact sum	PASS
TOTAL @94	1.0468061e-6	$> 1e-6$	same function	FAIL
Boundary	95 minimal	adjacent	verifier	CLOSED

The certified circuit uses only 24.18% of the control resources



One exact ratio
Merged groups, bond propagators and the
CNOT upper bound all scale by
 $11791/2851$ under the unchanged
compiler.

4.1357×
exact resource
reduction

−1.93M
CNOT upper
bound saved

Verification is layered for both live demos and deep audit

FAST · ~8 s

DEEP · 167.6 s

CLEAN CLONE

Reject corruption
Hashes, coverage,
group algebra, ledger
and integer resources.

Regenerate algebra
Rebuild every D4/D5
coefficient from the
31-stage formula.

Reproduce delivery
Fast verify, focused
tests and 11 SHA-256
checks all pass.

TAKEAWAY

4.1357418449x

— proved, not inferred

Same benchmark

95 pass · 94 fail

5x not claimed